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Executive Profile

A visionary and deeply technical executive leader with over 25 years of experience architecting and delivering market-defining technology for Fortune 100 corporations, global enterprises, and U.S. Federal Agencies. My career is defined by serving as a board-level advisor, Principal Investigator, and CTO for multiple high-growth ventures, leading them from ideation and capital allocation through to successful market launch, acquisition, and enterprise-scale deployment. As a hands-on leader (over 60%), I architect complex immutable & idempotent GenAI-Driven and quantum-ready solutions, ensuring they are governed by deep expertise in GRC frameworks (NIST, CMMC, SOC 2, HITRUST) to achieve FedRamp compliance and ATO certifications. My leadership drives digital transformation, modernizes the SDLC, uses advanced K8s orchestration and aligns technology with go-to-market strategies to solve high-stakes challenges and create decisive competitive advantages.

Core Competencies

AI & Machine Learning Leadership: Generative AI (LLMs, GPT, Llama), Retrieval-Augmented Generation (RAG), LangChain, Vector Databases, Quantum-Informed GRC, Computer Vision (CV), Natural Language Processing (NLP), MLOps, TensorFlow Extended (TFX), Semantic Intelligence, AI Product Strategy, AI Ethics, Evaluation, & Governance.

Cybersecurity, Governance, Risk & Compliance (GRC): NIST Cybersecurity Framework (CSF) & Risk Management Framework (RMF), CMMC L2/L3 Advisory, FedRamp Compliance & ATO Certification, SOC 2, HITRUST, PCI DSS, Well-Architected Frameworks (WAF), Shift-Left Security, GRC AI Platform Design, Quantum-Informed Risk Assessment.

Executive Strategy & Management: CTO-Level Leadership, Board-Level Advisory, Strategic Planning, P&L Management, Go-to-Market Strategy, Change Management, Digital Transformation, Federal Contracting (Principal Investigator), Thought Leadership, Building & Scaling Global Teams.

Enterprise Architecture & Cloud DevOps: AWS, Azure, Google Cloud Platform (GCP), Kubernetes, Docker, K8s, Advanced Containerization, Terraform, Infrastructure as Code (IaC), Microservices Architecture, CI/CD Pipelines (Jenkins, Azure DevOps), Serverless Computing, ITIL, SDLC Modernization, Ansible, Puppet, Chef.

Product & Solution Lifecycle Management: End-to-End Product Management, Solution Architecture, Go-to-Market Technical Strategy, SaaS Platform Development, Rapid Prototyping (MVP), Technical Due Diligence, Vendor & Partner Integration, Market Creation & Definition.

Professional Experience

Principal Architect & Strategic Consultant | AI, Quantum, & Security
2025 – Present

Public & Private Sector Consulting | Washington, *D.C. Metro Area*

Recruited to lead the full lifecycle of mission-critical AI and Enterprise Architecture solutions to modernize, transform, and migrate to State-of-the-Art (SOTA) AI-centered technical approaches. As a hands-on architect, I build secure, resilient systems from the ground up, leading development teams from initial design through to enterprise-scale implementation for a range of federal and commercial entities.

- Authored multiple agency's official Solution & Enterprise Architecture roadmaps, infused with Agentic AI. Established the strategic vision, ethical guidelines, and technical enterprise frameworks for secure AI adoption across complex federal enterprise and commercial landscapes.
- Serve as a key advisor to CTOs, CIOs, COOs and senior leaders in Federal Agencies, shaping policy on advanced cryptographic standards, enterprise AI governance, and "AI-driven shift-left" security integration into the SDLC.
- Architected and implemented resilient, secure AI workflows for the analysis of critical national security data, directly resulting in efficiency, de-duplication, automation, and visibility across the technical landscape.
- Lead technical deep-dive sessions and provided advisory on quantum-resistant cryptography and AI-Driven Well-Architected Frameworks and their implications for long-term automation and data security within the agency.

Distinguished Director of AI / Fractional CTO 2012 – 2025

Global Enterprise: Enterprise Technology Agency | San *Francisco, CA*

Served as a senior executive and principal consultant, leading a portfolio of high-impact client engagements as a **delegated CTO, Principal Architect, and strategic technology advisor**. Guided multiple organizations from early-stage startups to Fortune 500 enterprises through critical technology transformations, product launches, and security modernizations. Oversaw a multi-billion-dollar portfolio of projects, managed globally distributed teams of over 250 consultants, and was directly responsible for client success and technology delivery.

Key Client Engagements & Fractional Leadership Roles:

CTO | AI-Powered Federal Compliance & LegalTech Platform: Orchestrated the end-to-end development of a pioneering NLP and quantum-risk document intelligence engine. Architected a multi-cloud solution, led a team of 8 engineers, and scaled the

platform to process over 1 billion legal documents. Created a new service category in the federal compliance market, directly generating over 85 enterprise and international government clients.

CTO | TechStars-Backed AI Recruiting & HR Platform: Acted as CTO for a 40+ member team. Led the complete technical re-architecture of a Computer Vision-based HR Management platform, migrating to a scalable microservices framework on AWS using Bedrock and Sagemaker. Increased EEOC compliance accuracy by 30%, creating the core IP that secured strategic investment through the **TechStars Workforce Development Program** and led to a successful exit.

Director of Technology | Venture-Backed Beauty-On-Demand SaaS Platform: Spearheaded the migration of a fragmented static system to a secure, **Kubernetes-based microservices architecture on AWS**. Implemented a "shift-left" security strategy based on the Well-Architected Framework, creating rapid-deployment strategies that resulted in 10x deployment speed and achieving 99.99% uptime. Played a key technical role in due diligence, driving the company's security posturing and architectural philosophy, including the onboarding, training, and management of a 10-person full SDLC team.

Head of AI Product Development | Gartner Magic Quadrant-Leading Enterprise Software Company: Led the product and engineering teams that delivered an award-winning prediction pricing model. Drove the product strategy that earned an **AI Excellence Award** and repeat status as a **Leader in the Gartner Magic Quadrant**. Managed the full product lifecycle from ideation to deployment and maintenance, driving the cloud-architecture change management following ITIL practices and lean six sigma IT team management.

Enterprise Tech Consultant | Premier Atlassian Solutions Partner: Served as a core technical consultant to steer Atlassian adoption for defense agencies and national fintech providers focusing on thought leadership strategies, designing and implementing enterprise-scale DevOps and Agile workflows for complex client environments.

AI Technical Lead 2008 – 2012

HardMagic Corporation: Application Development | San *Francisco, CA*

Pioneered the company's strategic pivot into Artificial Intelligence. Led a dedicated team to build and deploy an innovative AI Engine, architecting a scalable enterprise Virtual Assistant and AI Customer Service solution from the ground up.

- Engineered an Omni-Channel Augmented CSR Virtual Agent experience with multipoint AI technology, including API Personality Integration for SMS, VoIP, Chat, Support Tickets, Email, and Social Media.

- Directed development across a polyglot stack including C#, C++, JavaScript, and Python to optimize performance, scalability, and user experience.
- Designed the underlying data models and conversational flow logic that became the core IP of the company's AI offerings.

Senior Software Developer 2004 – 2007

HardMagic Corporation | San *Francisco, CA*

Led development on the company's flagship SaaS products, focusing on scalability, security, and third-party integration for enterprise clients.

- Created a scalable Pedagogical Education SaaS Platform, engineering a responsive-layout, scalable MAAS + Juju code, and a PHP Drop-in Script.
- Architected and implemented custom-coded API integrations in a LAMP/LEMP stack with a wide range of third-party systems including WordPress, Moodle, Resource Space, Feng Office, and SMF Forum.
- Served as a lead developer for key initiatives with signature enterprise clients: **NIKE & The Disney Company**, focusing on hi-tech wearables, high-availability and secure content delivery systems.

Invented Methodologies & Patent-Pending Frameworks

Full-Lifecycle GRC & Security Automation Platform: Spearheaded the ideation, solution architecture, and rapid development of a market-disrupting GRC AI platform, taking it from concept to full-scale launch in just 12 months. Integrated with ServiceNow, Flexera, Kafka, and the US Defense EMASS system. The platform utilizes an AI Orchestrator to manage specialized agents for risk remediation (AI SRE) and automated incident response.

Advanced Agile Leadership Methodology: Collaborated directly with a key co-author of the original Agile Manifesto, to develop and implement a novel Agile Thought Leadership philosophy for the Enterprise & Defense Sector, focusing on cross-team functionality, performance management, distributed team management, lean agile philosophy, and integrating security and compliance into rapid development cycles.

A.N.D.R.O.I.D. Ops™ (AI Development Methodology): Pioneered a patent-pending proprietary Software Development Life Cycle (SDLC) tailored for complex Agentic SaaS products, integrating risk management, ethical considerations, and rapid iteration. This framework was adopted to lead a classified, cross-agency agile transformation initiative.

Focus-Tracker (Intelligent Agent Framework): Invented a sophisticated Agentic AI Project Management Platform. Utilized a Super Librarian AI Orchestration system focused on its novel approach of "Library -> Super-Librarian" agent architecture.

Technical Proficiency

Languages & Runtimes:

- **Rust**, **Go**, **TypeScript** (with Deno/Bun), **Python** (async-first), **Julia**, JavaScript, C#, Swift, Scala, Shell Scripting, SQL (SQLx, PostgREST), WASM/WebAssembly Runtimes
- **Quantum-Specific:** Qiskit, Cirq, OpenQASM

AI/ML & Data Science Ecosystem (MLEOps Focused):

- **Foundation Models/Generative AI Platforms:** OpenAI (GPT-5/next-gen), Anthropic (Claude-3.5/4/4.1), Google Gemini Advanced, Meta Llama (next-gen), Mistral AI, Amazon Bedrock, Microsoft Azure AI (CoPilot Studio, Cognitive Services)
- **AI/ML Frameworks & Libraries:** PyTorch (with TorchServe/Lightning), TensorFlow (with TFX), JAX, Hugging Face Transformers (Ecosystem), LangChain (Advanced RAG/Agent Orchestration), LlamaIndex, Ray (Distributed AI), Triton Inference Server, ONNX Runtime
- **Data Orchestration & Feature Stores:** Feast, Hopsworks, dbt (Data Build Tool), Airflow (Kubernetes-native), Prefect, Dagster
- **MLOps Platforms:** MLflow, Kubeflow, Vertex AI MLOps, SageMaker MLOps, Azure Machine Learning
- **Vector Databases & Retrieval:** Pinecone, Weaviate, Milvus, Qdrant

Cloud-Native & Distributed Platforms:

- **Hyperscaler Agnostic:** AWS (EKS Anywhere, Lambda, Fargate, S3, RDS, SageMaker), Azure (AKS, Functions, Cosmos DB, Azure AI, Container Apps), Google Cloud Platform (GKE, Cloud Run, BigQuery, Vertex AI)
- **Edge & IoT:** K3s, OpenYurt, Azure IoT Edge, AWS IoT Greengrass, Akri
- **Serverless Orchestration:** Kubernetes Event-driven Autoscaling (KEDA), Knative, Temporal, Cadence

DevOps, GitOps & Platform Engineering:

- **GitOps Orchestration:** Argo CD (Advanced Sync Strategies), Flux CD (Source Controller, Kustomize Controller), Crossplane (Universal Control Plane), Gitea
- **Kubernetes & Container Orchestration:** Kubernetes (K8s) (Advanced Operators, Custom Resources), Cilium (eBPF-based Networking/Security), Istio (Ambient Mesh), Linkerd, Envoy, OpenShift, AKS, EKS, GKE, Rancher (Fleet, Harvester, RKE2), Longhorn

- **Infrastructure as Code (IaC):** Pulumi (Multi-language, Policy-as-Code), Terraform (CDK for Terraform), Crossplane, CUE, AWS CDK, Azure Bicep
- **CI/CD & Developer Experience Platforms:** GitHub Actions (Self-hosted Runners, OIDC), GitLab CI/CD, Tekton, Spinnaker, Backstage (Developer Portals), Humanitec, Harness, Jenkins (Modern Declarative Pipelines)
- **Observability & AIOps:** Prometheus, Grafana, OpenTelemetry (Distributed Tracing, Metrics, Logs), Loki, Jaeger, Vector, Cortex, Thanos, Datadog (Integrated AIOps), Dynatrace, Cribl Stream
- **Service Mesh & API Management:** Istio (Ambient Mesh), Linkerd, Solo.io Gloo Mesh, Kong Gateway (Cloud-Native)
- **Security & Policy-as-Code:** Open Policy Agent (OPA) / Gatekeeper (Mutating/Validating Webhooks), Kyverno, Sigstore, Falco (Runtime Security), Trivy, Vault, Consul, SPIFFE/SPIRE, Cloud-Native Application Protection Platforms (CNAPP)

Quantum Computing & Technologies:

- **Quantum Development Frameworks:** Qiskit (IBM), Cirq (Google), PennyLane (Xanadu), Microsoft QDK (Q#)
- **Quantum Cloud Platforms:** IBM Quantum Experience, Amazon Braket, Azure Quantum, Google Cloud Quantum Engine
- **Quantum Algorithms:** QAOA, VQE, Shor's Algorithm, Grover's Algorithm, Quantum Machine Learning (QML)
- **Quantum Hardware Concepts:** Superconducting Qubits, Ion Traps, Neutral Atoms, Photonic Quantum Computing

Databases & Data Streaming:

- **Modern OLTP/NewSQL:** CockroachDB, YugabyteDB, Vitess (K8s-native sharding)
- **Vector Databases:** Pinecone, Weaviate, Milvus, Qdrant
- **Distributed SQL:** PostgreSQL (with extensions like TimescaleDB, Neon), MySQL (with Vitess/PlanetScale)
- **NoSQL:** MongoDB (Atlas/Realm), Redis (Enterprise/Cloud), Cassandra (Astra DB), DynamoDB, Couchbase
- **Graph Databases:** Neo4j (AuraDB), ArangoDB
- **Stream Processing:** Apache Kafka (Confluent Cloud), Apache Flink, Pulsar, RisingWave

- **Data Lakehouses:** Databricks Lakehouse, Delta Lake, Apache Iceberg, Apache Hudi

GRC & Cyber Security Operations:

- **Cloud-Native Security:** Cloud-Native Application Protection Platforms (CNAPP), Supply Chain Security (SLSA, Sigstore), Runtime Security (Falco), eBPF-based Security
- **Zero Trust & Identity:** SPIFFE/SPIRE, HashiCorp Vault, Okta, Auth0, Keycloak
- **XDR/SIEM:** SentinelOne (Singularity XDR), CrowdStrike Falcon, Splunk (Cloud/Enterprise Security), Elastic Security (SIEM/XDR), Microsoft Sentinel
- **Automated GRC:** Policy-as-Code (OPA, Kyverno), Compliance as Code (OpenControl), Security Chaos Engineering

Enterprise Ecosystems & Productivity (Cloud-Native & AI-Augmented):

- **AI-Enhanced Productivity:** Microsoft 365 Copilot, Google Workspace AI, Notion AI, Atlassian Intelligence (Jira, Confluence)
- **CRM/ERP:** Salesforce (AI Cloud), SAP S/4HANA (Cloud-Native), Microsoft Dynamics 365 (Copilot integration)
- **Service Management:** ServiceNow (Lightstep, Next-Gen ITSM), PagerDuty (AIOps)
- **Platform Engineering Tooling:** Backstage, Roadie, Internal Developer Platforms (IDP)

Academia & Research

- **SENIOR RESEARCHER | INSTAR LAB RESEARCH INSTITUTE (2020-2025):** Built an international team of Data Scientists and a virtual lab environment using Azure AI Document Intelligence, Data Labs, and DSVM Jupyter Hubs. Collaborated with an Educational Consortium including multiple Universities on interstellar research, quantum computing applications, and advanced DevOps methodologies.
- **ADJUNCT PROFESSOR, COMPUTER SCIENCE | UNIVERSITY OF CALIFORNIA, SANTA BARBARA (2005 - 2009):** Conducted academic research and taught courses with a focus on Human-Computer Interaction (HCI), Behavioral Engineering, and Applied Intelligence. Mentored students on projects that fused human thought and movement with AI systems.

Awards & Recognition

2020: Orchestrating AI Excellence – Spearheaded the development of **Divercity**, earning the prestigious **TechStars #1 AI HR App** award, revolutionizing talent acquisition with cutting-edge AI.

2021: Visionary Consulting Leadership – Recognized as the **#1 Atlassian Service Provider**, demonstrating unparalleled expertise and strategic impact as an Emerging Tech Consultant for Atlassian.

2022: Pioneering AI in Pricing – Led the program development for a **Gartner #1 AI Price Modeling App** with **PROS**, setting new industry benchmarks for Artificial Intelligence in pricing strategies.

2023: Redefining Legal Tech with AI – As AI CTO, architected **Contract Sent**, a **European Award-Winning Contract Intelligence Management Platform** that transformed legal processes.

2024: Innovating the Beauty Industry – Directed technology for **Priv**, achieving the **#1 Beauty on Demand iPhone App** with an outstanding **30,000+ 5-Star Ratings**, reflecting exceptional user experience and market success.

2025: Shaping the Future of Defense with AI – As an AI Enterprise Architect, drove **Award-Winning AI implementations within the Department of Defense (DoD)** alongside **Arch Systems Inc**, pushing the boundaries of national security technology.